Predict Then Verify: Closing the Loop on Curiosity-Driven Exploration to Resist the Noisy-TV Problem

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Abstract

Intrinsic motivation methods for exploration in reinforcement learning are vulnerable to the Noisy-TV problem, where stochastic stimuli trap agents into endlessly seeking novelty from non-informative distractors. We propose a *predict-then-verify* principle: instead of rewarding predicted information gain, reward only the measured decrease in model uncertainty after each transition. We instantiate this principle in two ways: Temporal PIG (TPIG), which uses KL-divergence, and a learning progress (LP) baseline using MSE. Both are minimal modifications to the Predicted Information Gain (PIG) algorithm. Evaluated on grid-world environments with up to 100 states, both TPIG and LP dramatically outperform PIG on distraction rate, exploration speed, and model accuracy, confirming that closed-loop verification of learning is the key to resisting stochastic distractors regardless of divergence measure.

1 Introduction

Reinforcement learning (RL) agents learn by taking actions in an environment and receiving rewards that guide future behavior [Sutton and Barto, 2018, Kaelbling et al., 1996]. When environmental rewards are sparse or absent, agents must generate their own *intrinsic* rewards to drive exploration [Aubret et al., 2019, Schmidhuber, 2010, Oudeyer et al., 2007]. Common intrinsic reward methods include count-based novelty [Tang et al., 2016, Bellemare et al., 2016], prediction-error curiosity [Pathak et al., 2017, Houthooft et al., 2016], and information-theoretic approaches [Little and Sommer, 2013].

A fundamental weakness of many intrinsic motivation methods is the *Noisy-TV problem* [Burda et al., 2019a]. Consider an environment containing a television with a remote control. Each button press displays a random, never-before-seen channel. Novelty-seeking agents become trapped, perpetually pressing the remote because each press yields a "new" state, even though no useful information about the environment is being learned.

Predicted Information Gain (PIG) [Little and Sommer, 2013] selects actions by estimating, via KL-divergence, how much each action would improve the agent's transition model. Although PIG is more principled than raw novelty-seeking, it still falls prey to the Noisy-TV: the stochastic transitions from button presses yield high predicted divergence, luring the agent back to the distractor.

The core issue is that PIG *predicts* information gain but never *confirms* that learning actually occurred. We propose a **predict-then-verify** approach: compute the predicted information gain both before and after incorporating a new transition, and use the *actual decrease* as a TD reward. We instantiate this principle as Temporal PIG (TPIG, using KL-divergence) and a learning progress baseline (LP, using MSE), and show that both resist the Noisy TV across environments ranging from 16 to 100 states.

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Many curiosity mechanisms are fundamentally open-loop: information gain, prediction error, and novelty signals can all remain high in stochastic environments even when no meaningful knowledge is acquired. We show that closing this loop by conditioning rewards on measured model improvement works under different divergence measures, indicating that the verification mechanism, not the specific curiosity signal, is the critical ingredient.

2 Related work

2.1 Intrinsic motivation for exploration

Intrinsic motivation methods let agents generate their own learning signals [Aubret et al., 2019, Schmidhuber, 2010], realized in several families of algorithms.

Count-based methods reward visiting novel states by tracking visitation counts. Bellemare et al. [2016] introduced pseudo-counts to unify count-based exploration with intrinsic motivation in high-dimensional spaces, and Tang et al. [2016] extended this approach using hash functions. However, count-based methods are directly vulnerable to the Noisy-TV problem: every channel change registers as a novel state, producing unbounded reward from a single distractor.

Prediction-error methods use the agent's inability to predict outcomes as a reward signal. Pathak et al. [2017] proposed the Intrinsic Curiosity Module (ICM), which trains a forward dynamics model and uses its prediction error as intrinsic reward. Houthooft et al. [2016] proposed VIME, which maximizes information gain about environment dynamics using Bayesian neural networks. Burda et al. [2019b] introduced Random Network Distillation (RND), which measures novelty as the prediction error of a randomly initialized fixed network. While more robust than count-based methods, Burda et al. [2019a] demonstrated that prediction-error agents can still be attracted to inherently unpredictable transitions such as those produced by a Noisy TV.

Information-theoretic methods, including PIG [Little and Sommer, 2013], estimate the expected change in the agent's world model using information-theoretic quantities such as KL-divergence. Little and Sommer [2013] showed that PIG-based exploration learns environment structure faster than reward-free baselines. Ménard et al. [2020] provided theoretical analysis of sample-efficient exploration strategies.

2.2 The noisy-TV problem

The Noisy-TV problem was identified by Burda et al. [2019a] and remains an open challenge; Ladosz et al. [2022] note that most intrinsic motivation methods remain susceptible [Kovac et al., 2020].

TPIG can also be viewed as an instantiation of *learning progress* [Oudeyer et al., 2007], which rewards reductions in prediction error over time. We make this connection explicit by also evaluating an MSE-based learning progress baseline (LP), allowing a direct comparison of divergence measures within the same TD framework. Both TPIG and LP operate directly on the agent's transition model without auxiliary networks, making them minimal modifications to PIG.

3 Background

3.1 Temporal difference learning

In temporal difference (TD) learning [Sutton and Barto, 2018], the agent maintains Q-values $Q(s, a)$ for each state-action pair [Watkins and Dayan, 1992] and updates them after each transition (s, a, s') using:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right], \tag{1}$$

where r is the reward, α is the learning rate, and γ is the discount factor [Sutton and Barto, 2018].

3.2 Predicted information gain (PIG)

PIG uses the KL-divergence [Kullback and Leibler, 1951] to estimate how much an agent's transition model would change if it took a particular action. Let $\hat{\Theta}_{sa\cdot}$ denote the agent's estimated

transition probabilities from state s under action a, and let $\hat{\Theta}_{sa\cdot}^{s\to s^*}$ denote the updated estimate after hypothetically observing a transition to s^*. The predicted information gain is:

$$\mathrm{PIG}(s,a) := \sum_{s^*} \hat{\Theta}_{sas^*}\, D_{\mathrm{KL}}\Big(\hat{\Theta}_{sa\cdot}^{s\to s^*} \parallel \hat{\Theta}_{sa\cdot}\Big), \tag{2}$$

where D_{KL} is the KL-divergence. PIG selects the action a maximizing this quantity. As data accumulates, $\hat{\Theta} \to \Theta$:

$$\lim_{n\to\infty} D_{\mathrm{KL}}(\Theta_{sa\cdot} \| \hat{\Theta}_{sa\cdot}) = \lim_{n\to\infty} \sum_{s'} \Theta_{sas'} \log_2 \frac{\Theta_{sas'}}{\hat{\Theta}_{sas'}} = 0,$$

since each ratio $\Theta_{sas'}/\hat{\Theta}_{sas'} \to 1$ and $\log_2(1) = 0$.

3.3 The noisy-TV problem

When the environment contains a Noisy TV, the stochastic channel changes produce transitions to many distinct states, each with high predicted information gain. PIG's action selection thus repeatedly favors the TV because the weighted average over all possible channel outcomes remains large, even though the transitions carry no useful structural information about the environment.

4 The predict-then-verify principle

TPIG modifies PIG by measuring the *actual* information gained after each transition rather than relying on predictions alone. After observing transition (s, a, s'), TPIG computes the predicted information gain both before and after updating the internal model:

$$r_{\mathrm{TPIG}}(s,a,s') := D_{\mathrm{KL}}\big(\hat{\Theta}_{sa\cdot}^{s\to s',\,t-1} \parallel \hat{\Theta}_{sa\cdot}^{t-1}\big) - D_{\mathrm{KL}}\big(\hat{\Theta}_{sa\cdot}^{s\to s',\,t} \parallel \hat{\Theta}_{sa\cdot}^{t}\big), \tag{3}$$

where superscripts $t-1$ and t denote the model before and after incorporating the observed transition. If genuine learning occurred, the remaining information decreases and the reward is positive. Both KL-divergence evaluations reuse PIG's existing computation, so TPIG adds only one additional divergence calculation per step. This reward updates Q-values via standard TD learning (1), with $\alpha = 0.9$ and $\gamma = 0.9$. Q-values are initialized to 1, providing an optimistic prior that encourages exploration of all actions before learned rewards dominate. TPIG selects the action with the highest Q-value:

$$\mathrm{TPIG}(s) = \arg\max_a Q(s,a).$$

4.1 The closed-minded model problem

After a single observation of (s, a, s'), PIG records that transition as 100% probable. On subsequent visits, the predicted and observed distributions are identical, the KL-divergence returns 0, and PIG considers the action fully learned, even if the true transition is stochastic. This "closed-minded" behavior causes PIG to neglect areas visited only once.

Both PIG and TPIG address this by initializing the total observed transition count to 1 rather than 0, equivalent to a uniform Dirichlet prior. This ensures the KL-divergence always indicates remaining information. As observations accumulate, the prior is overwhelmed and the model converges.

4.2 Handling infinities

When $\hat{\Theta}_{sa\cdot}$ is empty (no transitions observed), the KL-divergence returns infinity. These values are replaced with 0 in the reward calculation. Negative rewards (which can occur when discovering a new state temporarily increases uncertainty) are clamped to 0 to avoid penalizing exploration. Similarly, any Q-values that are infinite are normalized to 0 before the TD update is applied.

4.3 Why TPIG resists the noisy TV

PIG predicts high information gain from the TV because the model would change substantially. However, TPIG recomputes the predicted gain after the update and finds it is still high: each

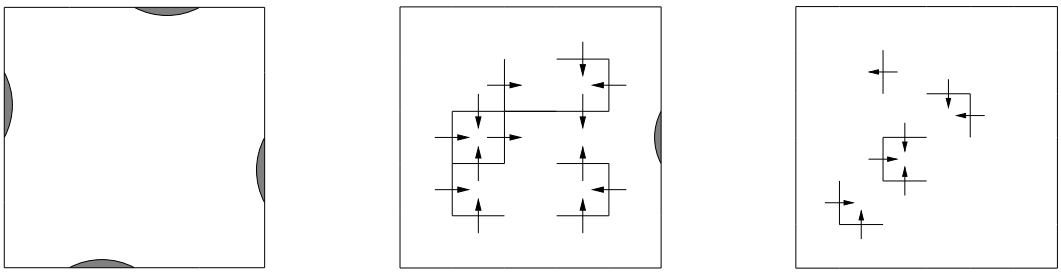


Figure 1: Grid-world simulations. Semi-circles are Noisy TVs; arrows indicate one-way walls. Left: 4-TV (4×4). Center: 1-TV (5×5). Right: No-TV (6×6).

observation barely reduces the remaining uncertainty because the TV states are so numerous. The reward is near zero, and TPIG's Q-values for TV-related actions decay. More precisely, for a stochastic transition with k equally likely outcomes, the n-th observation reduces predicted information gain by only $O(1/n)$, so the reward vanishes. The same argument applies to LP's MSE reward. Structured transitions produce larger, persistent reductions under either measure.

In summary, PIG takes the action it *predicts* will teach it the most. Both TPIG and LP take the action *experienced* to yield the most actual learning, a crucial distinction that makes the predict-then-verify approach robust to stochastic distractors.

4.4 Epsilon-greedy variants

We also evaluate ϵ-greedy versions of both algorithms: EPIG and ETPIG. With probability ϵ (set to 10% unless otherwise noted), the agent selects a random action instead of the greedy choice.

5 Experimental setup

5.1 Grid-world simulations

We test on four grid-world environments of increasing size (Figure 1):

- **4-TV** (4×4): Multiple Noisy TVs.
- **1-TV** (5×5): A single Noisy TV.
- **No-TV** (6×6): No distractors (control).
- **10x10-2TV** (10×10): Two Noisy TVs with internal walls creating corridors and rooms (100 states).

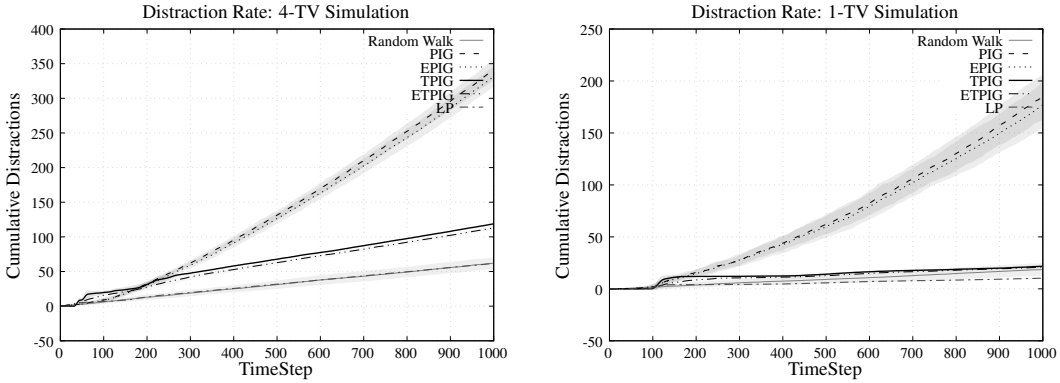
In each environment, actions move the agent in one of four cardinal directions. Movement succeeds with 90% probability; with 10% probability the agent remains in place. Walking into a wall always leaves the agent stationary. Activating a Noisy TV increments an unbounded noise variable in the state, so each activation produces a unique state, simulating a TV with an inexhaustible supply of channels. Importantly, the noise variable does not affect the actual transition dynamics: all TV states share the same transition probabilities regardless of the current channel. The noise is therefore a pure distractor: it inflates the apparent state space without adding any learnable structure.

All agents start in position 5. Each episode runs for 1000 time steps (3000 for the 10x10-2TV environment) with no goal state. Every policy is run for 100 independent trials per simulation. The source code and simulation data are publicly available.[1]

5.2 Metrics

- **Distraction rate**: Cumulative Noisy-TV activations.
- **Learning rate**: Number of undiscovered state transitions remaining.

[1] Repository URL withheld for double-blind review.



(a) 4-TV simulation (b) 1-TV simulation

Figure 2: Distraction rate. Shaded regions show ± 1 standard deviation over 100 trials.

- **Model accuracy**: KL-divergence between the agent's model and the true transition probabilities (excluding undiscovered transitions, which would contribute infinity).

5.3 Policies compared

Six policies are evaluated: Random Walk (uniform random baseline), PIG, EPIG (ϵ-greedy PIG), TPIG, ETPIG (ϵ-greedy TPIG), and LP (Learning Progress, a baseline that uses the decrease in prediction MSE rather than KL-divergence as its TD reward, using the same Q-learning framework as TPIG).

6 Results

6.1 Distraction rate

Figures 2a and 2b show cumulative distractions. PIG is heavily distracted, with its count growing steadily. Random Walk is distracted least since it has no bias toward any action. TPIG is initially distracted at a rate comparable to PIG but sharply reduces after a brief learning period. In the 1-TV environment, TPIG approaches Random Walk; in the 4-TV environment, the improvement is less dramatic but TPIG still substantially outperforms PIG. By the final time step, TPIG reduces distractions by 65% in the 4-TV environment and 88% in the 1-TV environment. The ϵ-greedy variants show reduced distraction as random action selection helps escape loops.

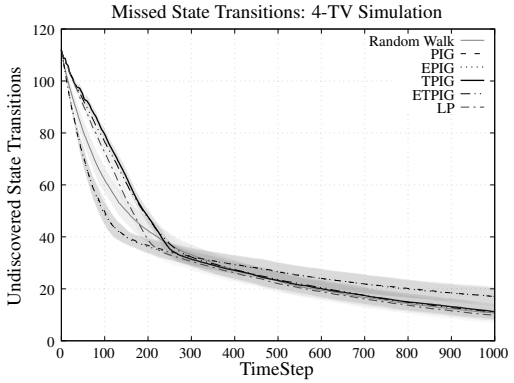
6.2 Learning rate

Figures 3a, 3b, and 3c show undiscovered transitions over time. PIG initially explores faster than Random Walk in the 1-TV and No-TV environments, but in the 4-TV environment it becomes trapped by the TVs and ultimately performs worse than Random Walk. TPIG starts slower but consistently surpasses both PIG and Random Walk after an initial learning phase, with the crossover occurring earlier in smaller environments.

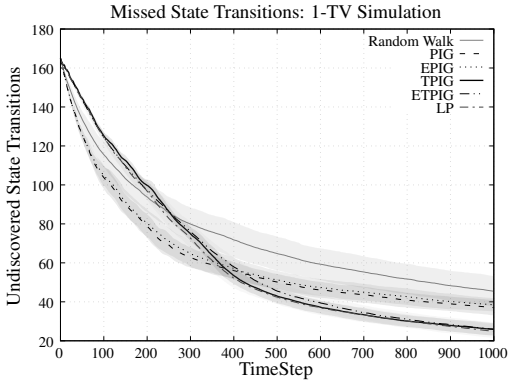
6.3 Model accuracy

Figures 4a, 4b, and 4c show KL-divergence between each agent's model and the true model. Undiscovered transitions are excluded (they would contribute infinity), so this metric must be evaluated jointly with the learning rate.

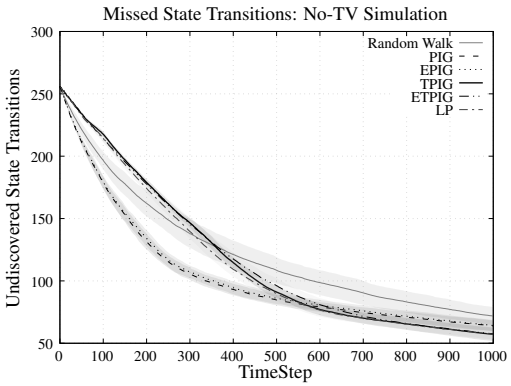
TPIG achieves lower KL-divergence than both PIG and Random Walk across all environments while having discovered more transitions, indicating both faster and more accurate learning. In the No-TV control (Figure 4c), TPIG still performs best, confirming that its advantage extends beyond noisy settings.



(a) 4-TV simulation



(b) 1-TV simulation



(c) No-TV simulation

Figure 3: Missed state transitions.

6.4 Scaling to a larger environment

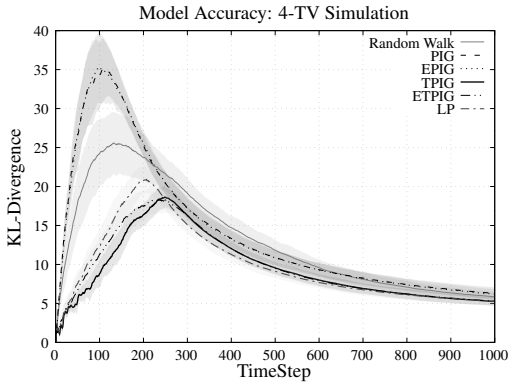
To test scaling, we evaluated all six policies on a 10×10 grid (100 states) with internal walls and two Noisy TVs (Figure 5). TPIG reduces distractions by 82% relative to PIG and achieves the best model accuracy. The two-phase behavior persists, with the crossover at approximately step 600, consistent with the linear scaling predicted from smaller environments.

6.5 Effect of epsilon

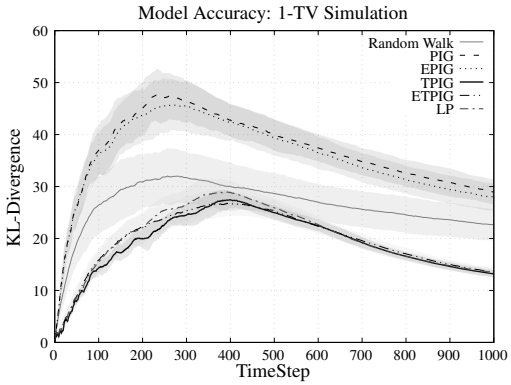
We tested ϵ in increments of 20%. Higher ϵ reduces distractions in the 4-TV simulation, but in 1-TV and No-TV, lower ϵ yields better long-term ETPIG performance, since TPIG's learned avoidance makes heavy randomization counterproductive. In the 1-TV ETPIG results (not shown), we observed a pivot around step 100 where all ϵ values shift from exploration to directed TV avoidance. Pure TPIG ($\epsilon = 0$) exhibits early artifacts smoothed by even small ϵ values.

6.6 Summary of results

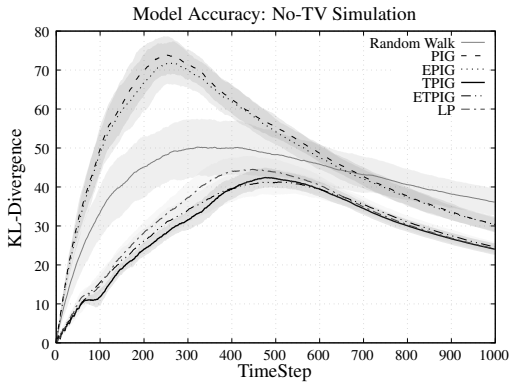
Table 1 reports final metric values averaged over 100 trials. TPIG and LP both dramatically outperform PIG on all metrics across all four environments. LP reduces distractions by 82–94% relative to PIG and achieves the lowest distraction counts, while TPIG achieves the best model accuracy; both explore faster than PIG. ETPIG and TPIG perform similarly, confirming that TPIG's learned avoidance eliminates the need for randomization.



(a) 4-TV simulation

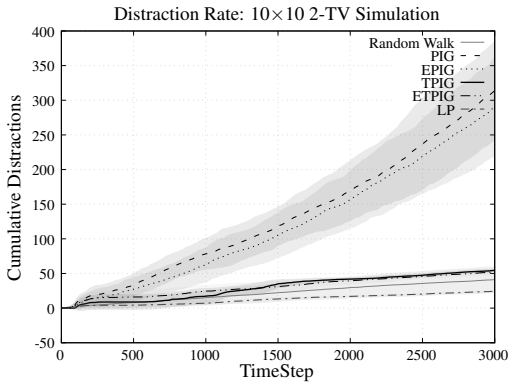


(b) 1-TV simulation

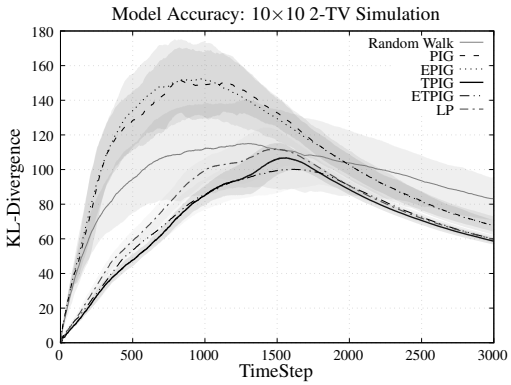


(c) No-TV simulation

Figure 4: Model accuracy (KL-divergence).



(a) Distraction rate



(b) Model accuracy

Figure 5: Results on the 10×10 2-TV environment (100 states, 3000 steps).

Table 1: Final metric values (mean over 100 trials). Best value in each column is **bold**. Lower is better for all metrics. Small environments run for 1000 steps; 10x10-2TV for 3000.

Environment	Policy	Distractions	Missed	KL-div
4-TV	Random Walk	62.0	10.7	5.86
	PIG	341.0	17.1	6.25
	EPIG	331.8	16.8	6.25
	TPIG	118.7	11.2	5.33
	ETPIG	112.8	11.2	5.22
	LP	**61.6**	**9.9**	**5.22**
1-TV	Random Walk	18.6	45.5	22.56
	PIG	184.2	37.0	28.86
	EPIG	175.6	39.2	28.32
	TPIG	21.7	25.9	**13.21**
	ETPIG	21.1	26.2	13.47
	LP	**10.2**	**24.9**	13.38
No-TV	Random Walk	**0.0**	71.9	36.12
	PIG	**0.0**	63.9	30.38
	EPIG	**0.0**	63.2	31.03
	TPIG	**0.0**	**57.2**	**24.01**
	ETPIG	**0.0**	57.7	24.59
	LP	**0.0**	**57.2**	24.24
10x10 2-TV	Random Walk	41.1	195.6	83.10
	PIG	313.6	179.0	67.50
	EPIG	277.4	178.5	67.10
	TPIG	54.5	161.1	**58.70**
	ETPIG	52.1	161.0	60.30
	LP	**24.2**	**154.6**	59.80

7 Discussion

By conditioning rewards on the measured decrease in model uncertainty, the predict-then-verify approach transforms exploration into a closed-loop feedback system. The Noisy TV generates large predictions but negligible actual learning, so Q-values for TV-related actions naturally decay under both TPIG and LP. Because the reward converges to zero as the model approaches the true distribution, both methods inherit TD convergence guarantees [Sutton and Barto, 2018], unlike open-loop curiosity methods that can be permanently misled by stochastic stimuli.

7.1 Learning dynamics

Both TPIG and LP exhibit two-phase behavior: initial broad exploration while accumulating data, followed by directed exploration once Q-values stabilize. PIG may outperform them during the first phase, but both surpass PIG on all metrics in the second. This pattern is consistent across all 100 trials, as the narrow error bands confirm.

The duration of this initial learning phase scales with environment size. In the 4×4 4-TV environment, the crossover occurs around time step 200; in the 5×5 1-TV environment, around time step 300; in the 6×6 No-TV environment, around step 400; and in the 10×10 10x10-2TV environment, around step 600. This approximately linear scaling suggests that TPIG's startup cost is predictable and proportional to the number of state-action pairs, representing 20–40% of the episode length across the environments tested. Table 2 shows a sensitivity analysis of α and γ on the 1-TV environment for both TPIG and LP. For TPIG, high α with low γ causes instability (521 distractions at α=0.9, γ=0.1), but all configurations with $\gamma \geq 0.5$ outperform PIG. LP is far more robust: it outperforms PIG in *every* parameter combination, including the setting that is catastrophic for TPIG. This confirms that the predict-then-verify mechanism is robust to hyperparameter choice, and that MSE is inherently more stable than KL-divergence for this application.

Table 2: Sensitivity to α and γ on the 1-TV environment (mean over 100 trials at step 999). PIG baseline: 184 distractions, 37 missed, 28.9 KL-div.

		$\gamma = 0.1$			$\gamma = 0.5$			$\gamma = 0.9$		
		Dist	Miss	KL	Dist	Miss	KL	Dist	Miss	KL
TPIG	α=0.1	21.7	34.2	24.3	19.0	31.3	21.5	27.1	32.7	16.4
	α=0.5	202.7	33.9	22.5	20.7	26.3	13.8	22.2	26.8	13.2
	α=0.9	521.5	41.6	29.7	107.6	28.0	14.4	**21.9**	26.2	13.2
LP	α=0.1	18.7	33.6	24.1	15.9	30.5	21.5	14.4	30.1	17.3
	α=0.5	13.2	29.2	17.6	10.9	26.3	13.7	10.9	26.4	13.2
	α=0.9	20.4	29.2	16.2	**10.2**	**26.1**	**13.1**	10.4	**25.9**	**13.1**

7.2 The role of epsilon

In environments with many distractors, higher ϵ helps both EPIG and ETPIG escape loops. In cleaner environments, lower ϵ yields better ETPIG performance because TPIG already learns to avoid distractors; additional randomization only slows convergence. For PIG-based methods, higher ϵ is consistently helpful since PIG lacks any mechanism to learn from its mistakes. LP was tested only in greedy mode; its strong distraction avoidance (Table 1) suggests it may benefit less from ϵ-greedy randomization than PIG does.

7.3 Comparison with other approaches

Unlike RND [Burda et al., 2019b] or ICM [Pathak et al., 2017], neither TPIG nor LP requires auxiliary networks, and both operate without any prior knowledge of which states are distractors or how the environment is structured. The LP baseline validates the generality of the predict-then-verify principle: replacing KL-divergence with MSE preserves noise resistance and yields even lower distraction rates, while both achieve comparable exploration speed and model accuracy. The choice of divergence measure is secondary to the closed-loop verification mechanism, and practitioners may prefer MSE for its simplicity and stronger distraction avoidance.

7.4 Limitations

The current study has several limitations. First, although we test on environments up to 100 states, they remain tabular. Scaling to continuous state spaces would require function approximation for the transition model and Q-values. Second, our unbounded noise model (infinitely many unique TV states) represents a worst case for novelty-based methods; real-world distractors with finite state spaces would be easier to overcome. Third, we do not combine TPIG's intrinsic reward with extrinsic rewards for goal-directed tasks. The $\pm 1\sigma$ bands in all figures confirm that the reported results are statistically robust.

8 Conclusion and future work

We introduced the predict-then-verify principle for curiosity-driven exploration and demonstrated it through two instantiations: TPIG (KL-divergence) and LP (MSE). Both dramatically outperform PIG across four grid-world environments ranging from 16 to 100 states and from zero to four distractors, with LP achieving the strongest distraction resistance and TPIG the best model accuracy. The convergence of results across divergence measures confirms that the key mechanism is closed-loop verification of learning, not the specific metric used; the choice between KL-divergence and MSE becomes a tunable design parameter rather than a fundamental algorithmic decision.

Future directions include scaling to continuous environments with function approximation, combining intrinsic and extrinsic rewards, and investigating whether a weighted combination of KL-divergence and MSE rewards can achieve both LP's distraction resistance and TPIG's model accuracy. The principle's simplicity and representation-independence make it a promising candidate for integration into deep RL curiosity methods [Pathak et al., 2017, Burda et al., 2019b].

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